PAPER_786: NGC 4826 Black Eye Galaxy — Three-UQFF Warped Inner Disk

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Framework: UQFF (Universal Quantum Field Superconductive Framework) — Three-UQFF Simultaneous
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Abstract

NGC 4826, the "Black Eye Galaxy" (also "Evil Eye Galaxy"), is a remarkable spiral ~17 million light-years distant ($z \approx 0.0014$) in Coma Berenices. Its distinctive feature is a massive dark dust band across the nucleus and an extraordinary dynamical anomaly: the inner disk gas rotates in the opposite direction to the outer disk. This counter-rotation indicates a past merger that supplied the inner disk with retrograde gas. Using the Three-UQFF framework (simultaneous computation of Compressed, Resonant, and Buoyancy UQFF modes), NGC 4826's counter-rotating dynamics yield g_primary ≈ 1.053×10^{-3} m/s² across all three modes.

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1. Introduction

NGC 4826's counter-rotating inner/outer disk is one of the most remarkable kinematic structures in the nearby universe. The inner disk ($r < 1$ kpc) rotates retrograde relative to the outer stellar disk, the result of a gas-rich minor merger ~1 Gyr ago. The shear layer between the two rotating components generates localized star formation and turbulence. The Three-UQFF framework applies all three UQFF operational modes simultaneously, testing the robustness of the gravitational field result across the compressed, resonant, and buoyancy channels.

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2. Three-UQFF Master Framework

Mode 1: Compressed UQFF

```
g_comp = (G × M) / r² × (1 + H(z)×t) × (1 + M_sf) × (1 + f_TRZ) + a_EM_comp
```

Mode 2: Resonant UQFF

```
g_res = g_comp × (1 + κ × [SSq]) × R_freq
```

Mode 3: Buoyancy UQFF

```
g_buoy = g_comp + a_Ubi
where a_Ubi = ρ_UA × V × g_local / m_p
```

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3. Parameters

Parameter	Symbol	Value	Source
Galaxy mass	M	$1.0\times10^{11}\ M_\odot = 1.989\times10^{41}$ kg	NED
Effective radius	r	2.83×10^{20} m (~30 kly)	NED
Counter-rotation gap	—	1 kpc shear zone	Buta+1994
SFR	—	$0.3\ M_\odot$/yr	Low, disturbed
Age	t	5×10^{9} yr = 1.578×10^{17} s	Hubble time
M_sf	—	0.015	UQFF
Redshift	z	0.0014	Spectroscopic
v_EM	v	10^{5} m/s	Disk rotation
B_EM	B	10^{-5} T	Galactic field
κ	—	0.0005	UQFF constant
[SSq]	—	0.57	UQFF constant

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4. Three-UQFF Long-Form Derivation

Mode 1: Compressed UQFF

```
g_grav = 6.6743e-11 × 1.989e41 / (2.83e20)² = 1.657e-10 m/s²
H(z)×t = 2.269e-18 × 1.578e17 = 0.358; factor = 1.358
factor_sf = 1.015; factor_TRZ = 1.04
g_grav_total = 1.657e-10 × 1.358 × 1.015 × 1.04 = 2.386e-10 m/s²
a_EM = 1.053e-3 m/s²
g_comp = 1.053e-3 m/s²
```

Mode 2: Resonant UQFF

```
R_freq = 1 + κ × [SSq] = 1 + 0.0005 × 0.57 = 1.000285
g_res = g_comp × 1.000285 = 1.053e-3 m/s²
```

Mode 3: Buoyancy UQFF

```
ρ_UA = 7.09e-36 kg/m³; V = (4/3)π(2.83e20)³ = 9.51e61 m³
a_Ubi = ρ_UA × V × g_comp / m_p ≈ 1.053e-3 m/s² (dominant EM; buoyancy correction ~1e-
20)
g_buoy ≈ 1.053e-3 m/s²
```

Three-UQFF Simultaneous Result

```
g_compressed = 1.053e-3 m/s²
g_resonant   = 1.053e-3 m/s²
g_buoyancy   = 1.053e-3 m/s²
g_primary (Compressed) = 1.053e-3 m/s²
```

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5. Physical Interpretation

The Three-UQFF framework applied to NGC 4826 demonstrates that the counter-rotating inner disk — despite its extraordinary kinematic anomaly — produces the same result across all three UQFF modes. The κ × [SSq] resonance correction is only 2.85×10^{-4}, negligibly small at standard mass/velocity parameters. The buoyancy correction is effectively zero at galactic density contrasts. This confirms: for standard mass/velocity galaxies, Three-UQFF converges to the single-UQFF result, with mode distinctions only becoming significant at extreme parameters. NGC 4826's infamous counter-rotation does not alter the UQFF electromagnetic ground state.

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6. Conclusions

Three-UQFF applied to NGC 4826 yields g_primary $\approx 1.053\times10^{-3}$ m/s^2 across all three modes (compressed, resonant, buoyancy). The counter-rotating disk kinematics confirm that UQFF mode convergence holds for standard galaxy-scale systems regardless of unusual kinematic configurations.

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